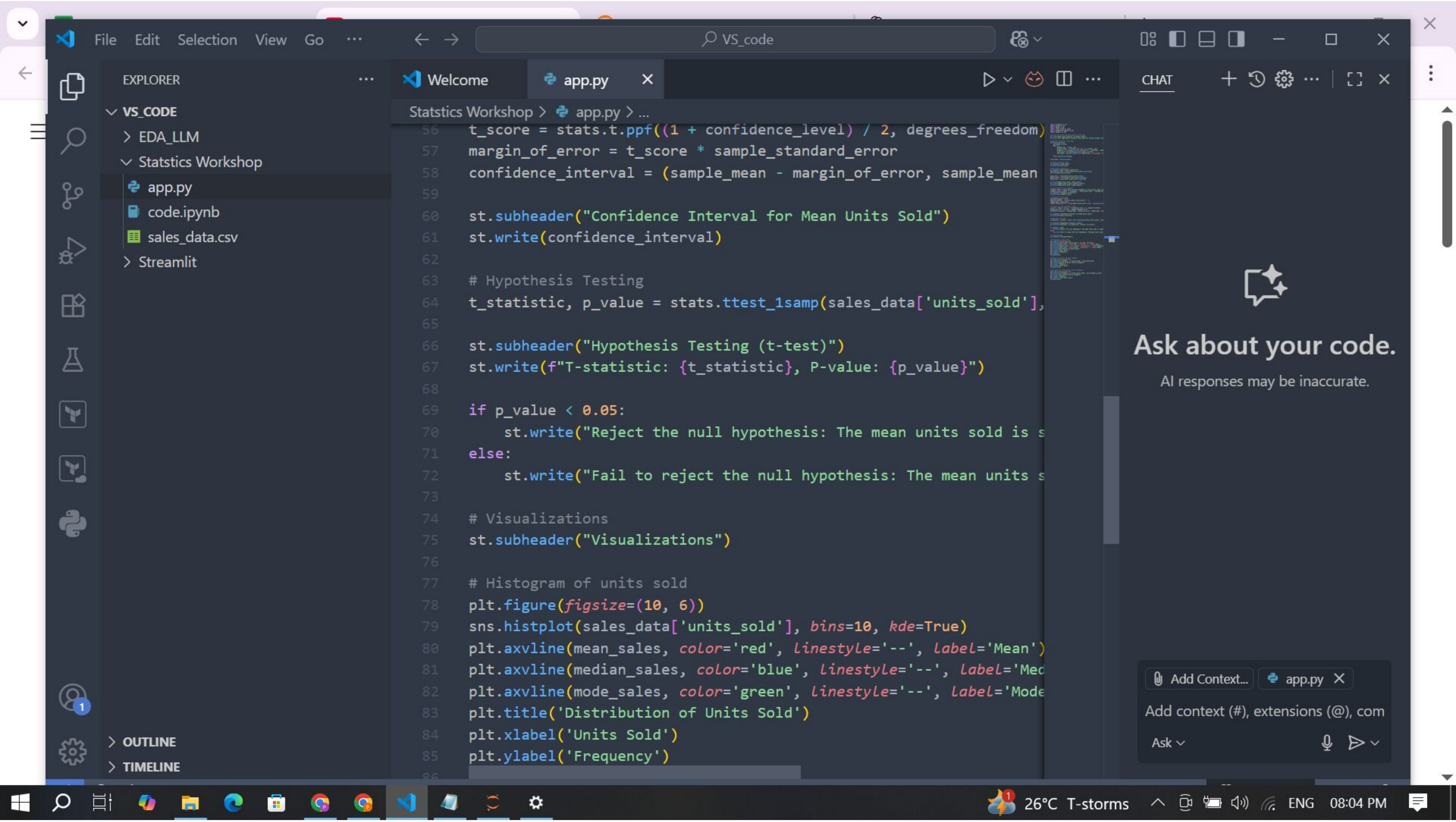
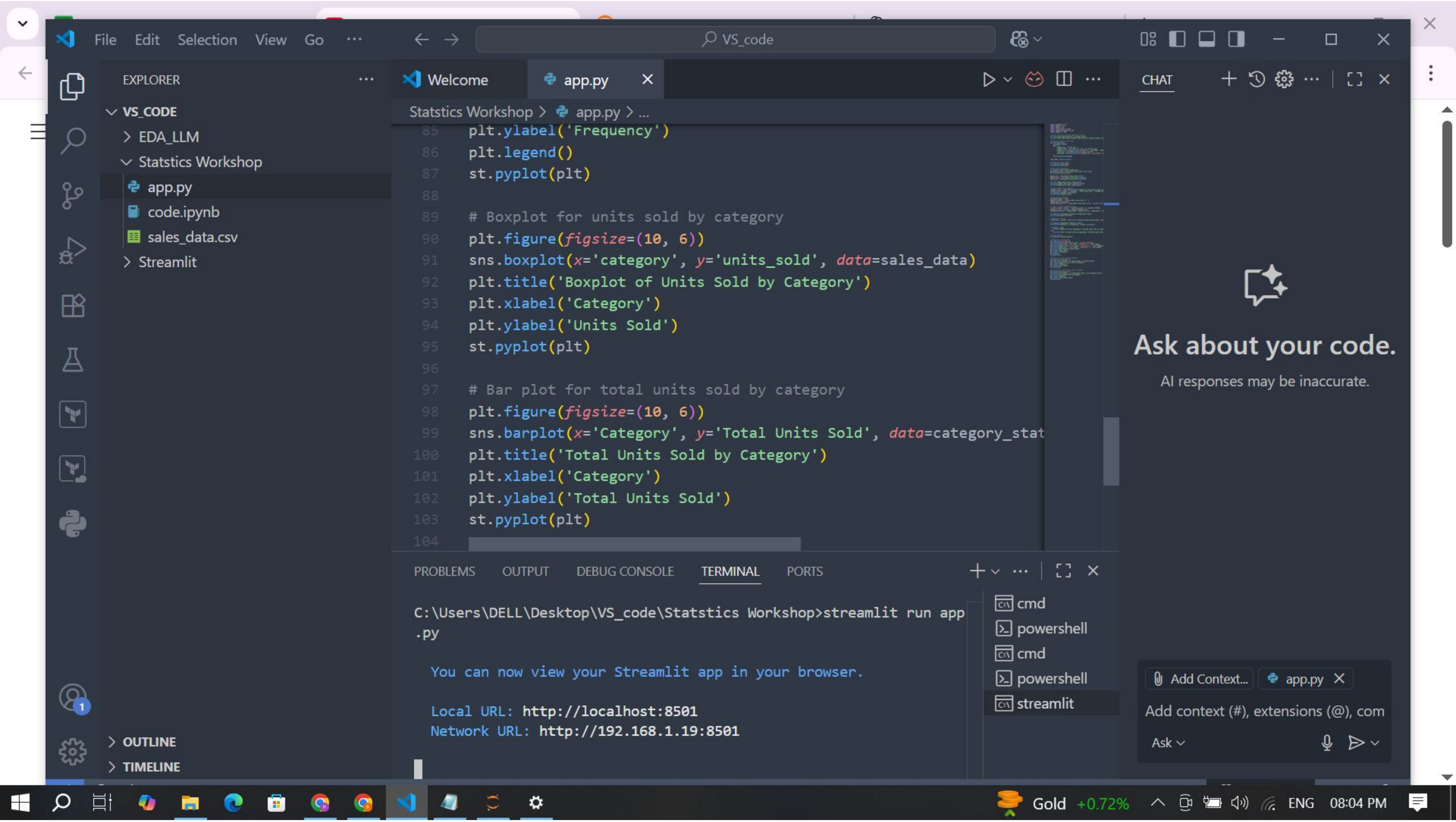
```python
import streamlit as st
import pandas as pd
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import seaborn as sns

# Set up the title and description of the app
st.title("Sales Data Analysis for Retail Store")
st.write("This application analyzes sales data for various product

# Generate synthetic sales data
def generate_data():
    np.random.seed(42)
    data = {
        'product_id': range(1, 21),
        'product_name': [f'Product {i}' for i in range(1, 21)],
        'category': np.random.choice(['Electronics', 'Clothing', '
        'units_sold': np.random.poisson(lam=20, size=20),
        'sale_date': pd.date_range(start='2023-01-01', periods=20,
    }
    return pd.DataFrame(data)

sales_data = generate_data()

# Display the sales data
st.subheader("Sales Data")
st.dataframe(sales_data)

# Descriptive Statistics
```

```python
st.subheader("Sales Data")
st.dataframe(sales_data)

# Descriptive Statistics
st.subheader("Descriptive Statistics")
descriptive_stats = sales_data['units_sold'].describe()
st.write(descriptive_stats)

mean_sales = sales_data['units_sold'].mean()
median_sales = sales_data['units_sold'].median()
mode_sales = sales_data['units_sold'].mode()[0]

st.write(f"Mean Units Sold: {mean_sales}")
st.write(f"Median Units Sold: {median_sales}")
st.write(f"Mode Units Sold: {mode_sales}")

# Group statistics by category
category_stats = sales_data.groupby('category')['units_sold'].agg(
category_stats.columns = ['Category', 'Total Units Sold', 'Average
st.subheader("Category Statistics")
st.dataframe(category_stats)

# Inferential Statistics
confidence_level = 0.95
degrees_freedom = len(sales_data['units_sold']) - 1
sample_mean = mean_sales
sample_standard_error = sales_data['units_sold'].std() / np.sqrt(l

# t-score for the confidence level
t_score = stats.t.ppf((1 + confidence_level) / 2, degrees_freedom)
```


File
Edit
Selection
View
Go
VS_code
EXPLORER
VS_CODE
EDA_LLM
Statstics Workshop
app.py
code.ipynb
sales_data.csv
Streamlit
OUTLINE
TIMELINE
Welcome
app.py
Statstics Workshop > app.py > ...
t_score = stats.t.ppf((1 + confidence_level) / 2, degrees_freedom)
margin_of_error = t_score * sample_standard_error
confidence_interval = (sample_mean - margin_of_error, sample_mean
st.subheader("Confidence Interval for Mean Units Sold")
st.write(confidence_interval)
# Hypothesis Testing
t_statistic, p_value = stats.ttest_1samp(sales_data['units_sold'],
st.subheader("Hypothesis Testing (t-test)")
st.write(f"T-statistic: {t_statistic}, P-value: {p_value}")
if p_value < 0.05:
    st.write("Reject the null hypothesis: The mean units sold is s
else:
    st.write("Fail to reject the null hypothesis: The mean units s
# Visualizations
st.subheader("Visualizations")
# Histogram of units sold
plt.figure(figsize=(10, 6))
sns.histplot(sales_data['units_sold'], bins=10, kde=True)
plt.axvline(mean_sales, color='red', linestyle='--', label='Mean')
plt.axvline(median_sales, color='blue', linestyle='--', label='Med
plt.axvline(mode_sales, color='green', linestyle='--', label='Mode
plt.title('Distribution of Units Sold')
plt.xlabel('Units Sold')
plt.ylabel('Frequency')
CHAT
Ask about your code.
AI responses may be inaccurate.
Add Context...
app.py
Add context (#), extensions (@), com
Ask
26°C T-storms
ENG
08:04 PM


File Edit Selection View Go ...
VS_code
EXPLORER
VS_CODE
EDA_LLM
Statstics Workshop
app.py
code.ipynb
sales_data.csv
Streamlit
OUTLINE
TIMELINE
Welcome
app.py
Statstics Workshop > app.py > ...
plt.ylabel('Frequency')
plt.legend()
st.pyplot(plt)
# Boxplot for units sold by category
plt.figure(figsize=(10, 6))
sns.boxplot(x='category', y='units_sold', data=sales_data)
plt.title('Boxplot of Units Sold by Category')
plt.xlabel('Category')
plt.ylabel('Units Sold')
st.pyplot(plt)
# Bar plot for total units sold by category
plt.figure(figsize=(10, 6))
sns.barplot(x='Category', y='Total Units Sold', data=category_stat
plt.title('Total Units Sold by Category')
plt.xlabel('Category')
plt.ylabel('Total Units Sold')
st.pyplot(plt)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
C:\Users\DELL\Desktop\VS_code\Statstics Workshop>streamlit run app.py
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://192.168.1.19:8501
cmd
powershell
cmd
powershell
streamlit
CHAT
Ask about your code.
AI responses may be inaccurate.
Add Context...
app.py
Add context (#), extensions (@), com
Ask
Gold +0.72%
ENG
08:04 PM

Sales Data Analysis for Retail Store

This application analyzes sales data for various product categories.

Sales Data

	product_id	product_name	category	units_sold	sale_date
0	1	Product 1	Home	25	2023-01-01 00:00:00
1	2	Product 2	Sports	15	2023-01-02 00:00:00
2	3	Product 3	Electronics	17	2023-01-03 00:00:00
3	4	Product 4	Home	19	2023-01-04 00:00:00
4	5	Product 5	Home	21	2023-01-05 00:00:00
5	6	Product 6	Sports	17	2023-01-06 00:00:00
6	7	Product 7	Electronics	19	2023-01-07 00:00:00
7	8	Product 8	Electronics	16	2023-01-08 00:00:00
8	9	Product 9	Home	21	2023-01-09 00:00:00
9	10	Product 10	Clothing	21	2023-01-10 00:00:00

Descriptive Statistics

	units_sold
count	20
mean	18.8
std	3.3023
min	13
25%	17
50%	18.5
75%	21
max	25

Mean Units Sold: 18.8

Median Units Sold: 18.5

Mode Units Sold: 17

Category Statistics

	Category	Total Units Sold	Average Units Sold	Std Dev of Units Sold
0	Clothing	21	21	None
1	Electronics	73	18.25	2.2174
2	Home	181	20.1111	3.7231
3	Sports	101	16.8333	2.7142

Confidence Interval for Mean Units Sold

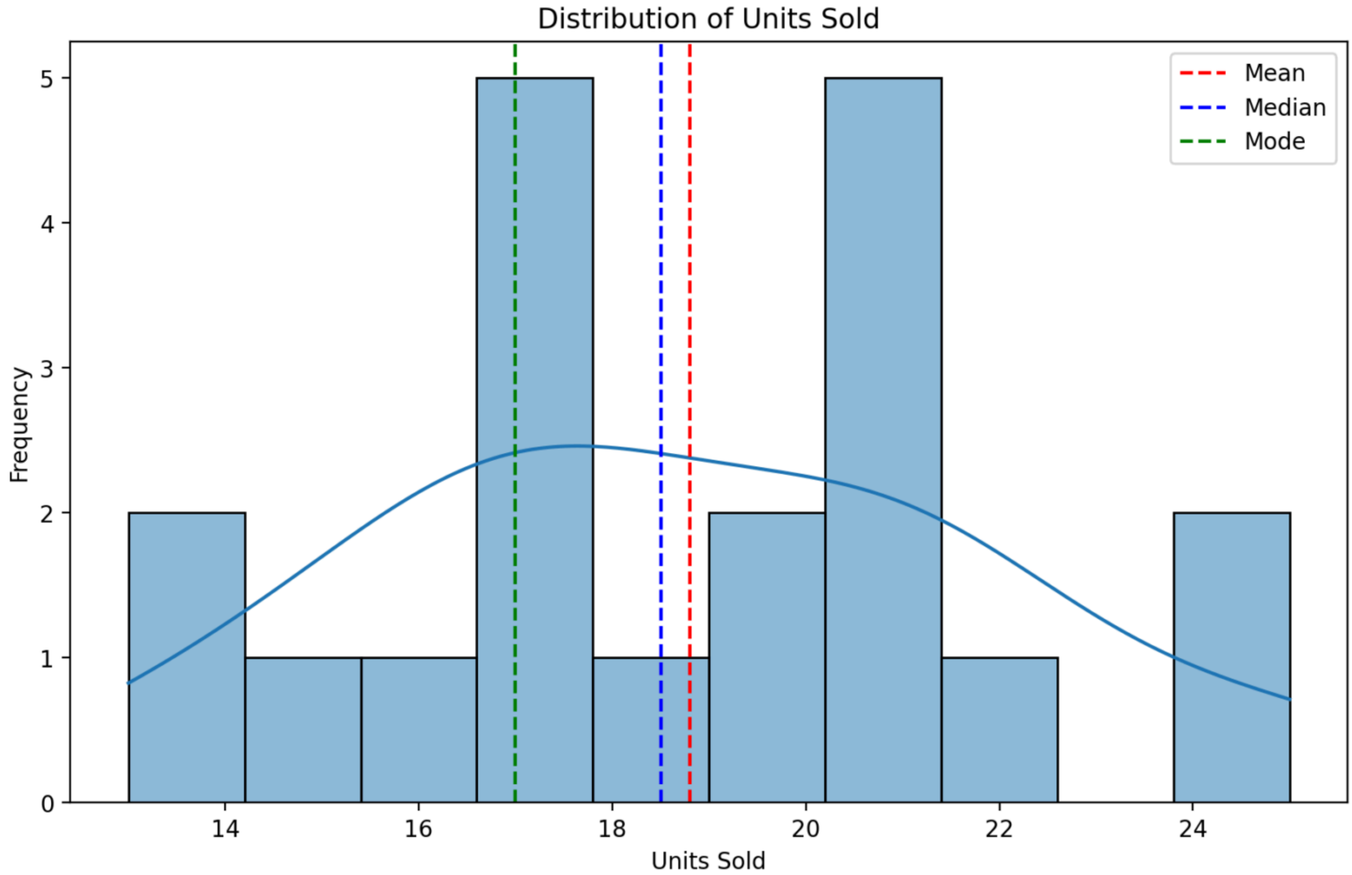
```
(np.float64(17.254470507823573), np.float64(20.34552949217643))
```

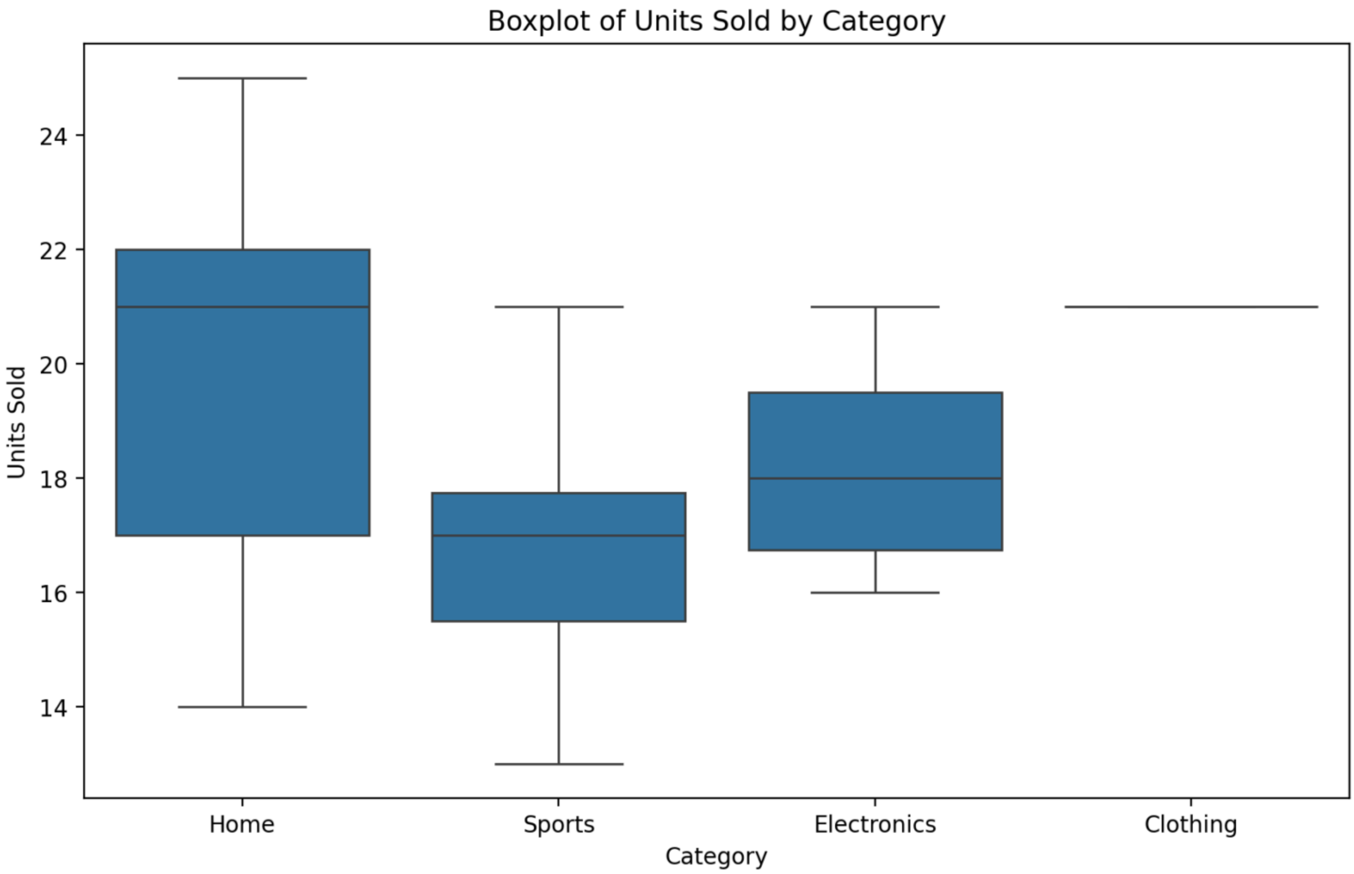
Hypothesis Testing (t-test)

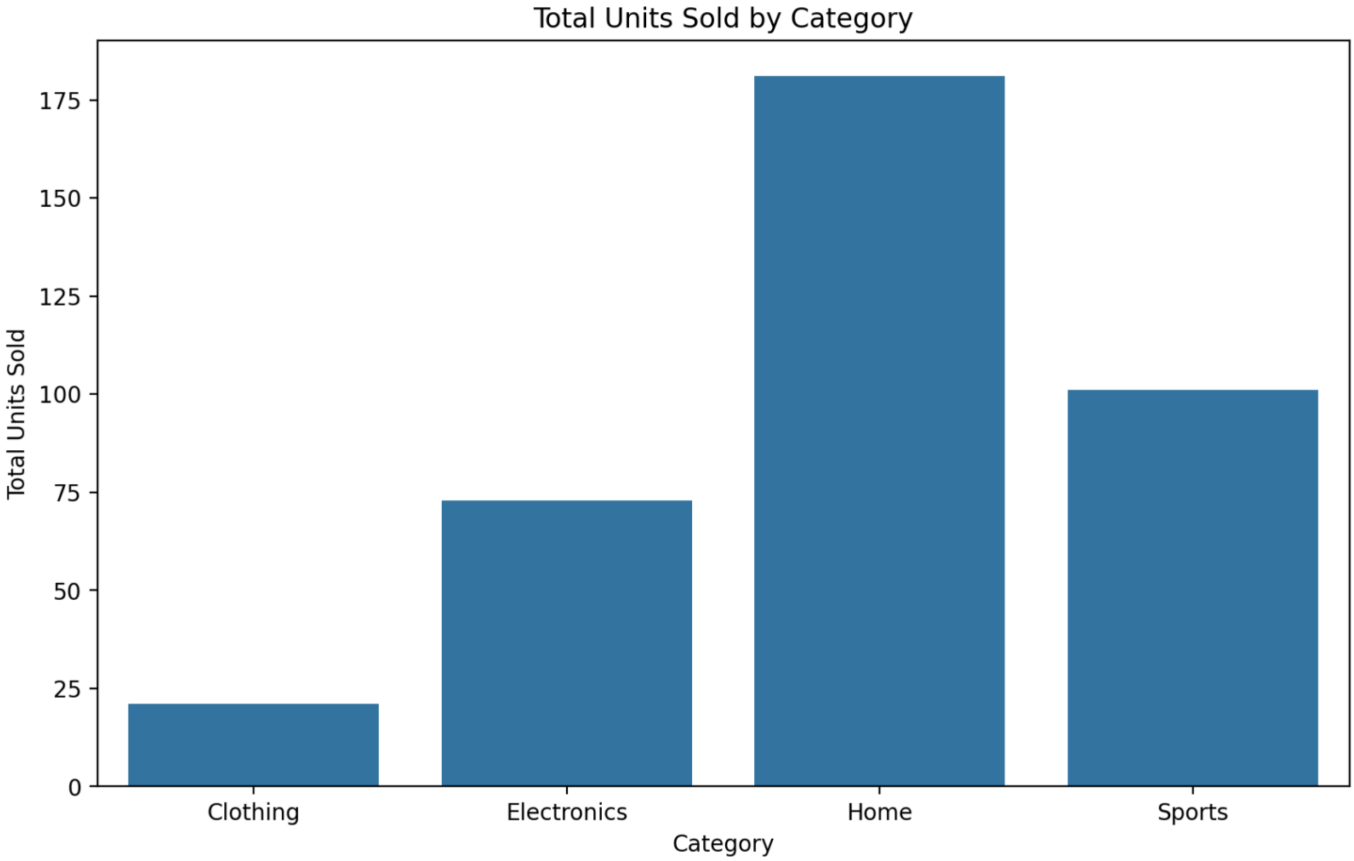
T-statistic: -1.6250928099424466, P-value: 0.12061572226781002

Fail to reject the null hypothesis: The mean units sold is not significantly different from 20.

Visualizations


Distribution of Units Sold
Mean
Median
Mode
Frequency
0
1
2
3
4
5
14
16
18
20
22
24
Units Sold


Boxplot of Units Sold by Category
Units Sold
14
16
18
20
22
24
Home
Sports
Electronics
Clothing
Category


Total Units Sold by Category
Total Units Sold
175
150
125
100
75
50
25
0
Clothing
Electronics
Home
Sports
Category